

Brent's Method

input : function f , bracket $[a, b]$ with $f(a) \cdot f(b) < 0$, tolerance ε_s

output: estimate b of the root

```
// b is the best estimate, c is on the other side of the root, a
  is the previous b
c ← a;   d ← b - a;   e ← d
repeat
  if  $f(b) \cdot f(c) > 0$  then
    // b and c no longer bracket the root; a does
    c ← a;   d ← b - a;   e ← d
  end
  if  $|f(c)| < |f(b)|$  then
    // make b the point with the smallest |f|
    a ← b;   b ← c;   c ← a
  end
  m ← (c - b) / 2                                // bisection step
   $\delta \leftarrow \varepsilon_s |b|$                   // absolute tolerance
  if  $|m| \leq \delta$  or  $f(b) = 0$  then
    return b
  end
  if  $|e| < \delta$  or  $|f(a)| \leq |f(b)|$  then
    d ← m;   e ← m                                // bisect
  else
    s ←  $f(b) / f(a)$ 
    if a = c then
      // only two distinct points: secant step
      p ← 2ms;   q ← 1 - s
    else
      // three distinct points: inverse quadratic
      interpolation
      q ←  $f(a) / f(c)$ ;   r ←  $f(b) / f(c)$ 
      p ←  $s(2mq(q - r) - (b - a)(r - 1))$ 
      q ←  $(q - 1)(r - 1)(s - 1)$ 
    end
    if p > 0 then q ← -q
    else p ← -p
    if  $2p < 3mq - |\delta q|$  and  $p < |eq/2|$  then
      // interpolated step stays in the bracket and shrinks
      it fast enough
      e ← d;   d ← p / q
    else
      d ← m;   e ← m                                // bisect
    end
  end
end
a ← b
if  $|d| > \delta$  then b ← b + d
else b ← b +  $\delta \text{sign}(m)$ 
until converged
return b
```